ENTRY CODE: MCS-T-R04A-017_3

Research Plan Instructions

A complete research plan is required and must accompany Checklist for Student (1A)

GLUCO-MATH: A Predictive Algorithmic System Utilizing Non-Linear Differential Models of Subcutaneous Insulin Absorption Integrated with Fuzzy Logic for Proactive Hypoglycemic Mitigation

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A. Rationale

Diabetes Mellitus is a chronic metabolic disorder characterised by hyperglycemia—persistently high blood glucose levels—resulting from defects in insulin secretion, insulin action, or a combination of both. This condition encompasses several categories, primarily Type 1 and Type 2 diabetes, which stem from distinct pathophysiological mechanisms: the autoimmune destruction of pancreatic beta cells or a progressive imbalance between insulin production and sensitivity. Without a precise balance between insulin and glucagon, glucose levels become skewed, leading to acute osmotic diuresis and long-term nonenzymatic glycation of proteins. This damage to small blood vessels eventually manifests as severe complications, including retinopathy, nephropathy, and neuropathy[1]. Despite the availability of purification and management strategies, diabetes remains a critical global health challenge, requiring precise blood glucose regulation to prevent life-threatening complications [2].

Amid the advancements in therapy, achieving tight glycemic control remains exceptionally difficult for patients. To manage this, the medical community has shifted

[1]Amit Sapra and Priyanka Bhandari, "Diabetes," in StatPearls (Treasure Island, FL: StatPearls Publishing, 2023), https://www.ncbi.nlm.nih.gov/books/NBK551501/.
[2]World Health Organization, "Diabetes," Fact sheets, updated April 5, 2024, https://www.who.int/news-room/fact-sheets/detail/diabetes.

toward automated insulin delivery systems—often called "closed-loop" systems or artificial pancreases—which aim to mimic the function of a healthy pancreas by continuously monitoring glucose levels and automatically adjusting insulin administration.

These systems matter because they offer the potential to significantly reduce the cognitive burden on patients, who otherwise must manually calculate and inject insulin multiple times a day. However, current automated systems often fail to simulate the complex, non-linear dynamics of human metabolism, leading to issues in blood glucose regulation[3]. Many existing automated insulin systems operate on a reactive principle; they analyze current blood glucose measurements to determine insulin dosing, rather than proactively anticipating future glycemic fluctuations based on historical data or external factors[4].

The limitation of conventional automated algorithms, such as Proportional-Integral-Derivative (PID) controllers, stems from their inability to handle significant intra-patient variability. The same patient may exhibit vastly different metabolic responses to identical insulin doses based on daily changes in diet, physical activity, and stress levels[5]. Furthermore, the requirement for precise carbohydrate counting in current closed-loop systems introduces significant control errors, as patients often struggle to provide accurate estimations, leading to substantial noise within the insulin delivery algorithm[6].

To address these limitations, this study proposes GLUCO-MATH, a hybrid predictive software system designed to merge the mathematical rigor of Non-Linear Differential Equations (NLDE) with the flexible decision-making of Fuzzy Logic. While NLDE models

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[3] J. L. C. B. De Farias and W. M. Bessa, "Intelligent Control with Artificial Neural Networks for Automated Insulin Delivery Systems," Bioengineering 9, no. 11 (2022): 664, https://doi.org/10.3390/bioengineering9110664.

[4] Melanie K. Bothe et al., "The Use of Reinforcement Learning Algorithms to Meet the Challenges of an Artificial Pancreas," Computational and Mathematical Methods in Medicine 2018 (2018): 6, https://doi.org/10.1155/2018/8504265.

[5] Mohammad R. Askari et al., "Multivariable Automated Insulin Delivery System for Handling Planned and Spontaneous Physical Activities," Journal of Diabetes Science and Technology 17, no. 6 (2023): 1457, https://doi.org/10.1177/19322968231204884.

[6] H. Lee et al., "The Effect of Carbohydrate Counting Accuracy on Postprandial Glucose Control Using a Closed-Loop Artificial Pancreas System," Journal of Diabetes Science and Technology 11, no. 6 (2017): 1112, https://doi.org/10.1177/1932296817711467.

provide precise physiological modeling, they struggle with the unpredictable, subjective inputs of daily life. Conversely, pure fuzzy logic systems handle these linguistic inputs well but lack the rigorous mathematical foundation for long-term physiological prediction. GLUCO-MATH bridges this gap by using a modified NLDE model for the 'physiology engine' and Fuzzy Logic for the 'decision-making engine.' This approach is essential for modeling complex, non-linear delays in subcutaneous insulin absorption and the metabolic oscillations caused by physiological changes like exercise or stress[7]. Complementing this, the integration of Fuzzy Logic using the scikit-fuzzy library addresses the inherent uncertainty of real-world human variables by employing linguistic rules and overlapping membership functions, rather than relying on precise numerical inputs, which allows for a more flexible and accurate approach to insulin dosage calculations[8]. By utilizing fuzzy membership functions, the system translates vague linguistic inputs—"high stress" or "moderate exercise"—into precise mathematical values.

The technical feasibility of this research is grounded in a Python programming environment, utilizing SciPy for the numerical integration of differential equations. To ensure patient safety during development, the py-mgipsim simulator will be used to generate virtual Type 1 Diabetes patients for algorithmic validation. GLUCO-MATH acts as a computational "middleman" that connects to real-time data sources. It utilizes the Nightscout cloud platform as a data bridge to pull glucose values via Abbott FreeStyle Libre 3 sensors. The software then produces a visual Prediction Horizon forecasting blood sugar trends 30–60 minutes in advance. This allows the controller to shift from reactive corrections to proactive prevention, automatically adjusting insulin delivery to mitigate hypoglycemia before it occurs.

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[7]Mohammad R. Askari et al., "Multivariable Automated Insulin Delivery System for Handling Planned and Spontaneous Physical Activities," Journal of Diabetes Science and Technology 17, no. 6 (2023): 1459, https://doi.org/10.1177/19322968231204884.
[8]Tajul Rosli Razak et al., "Python scikit-fuzzy: Developing a Fuzzy Expert System for Diabetes Diagnosis," IAES International Journal of Artificial Intelligence (IJ-AI) 13, no. 2 (2024): 2199, https://doi.org/10.11591/ijai.v13.i2.pp2197-2206.

Ultimately, GLUCO-MATH aims to deliver a personalized, closed-loop framework that increases patient safety, improves Time-in-Range (TIR), and significantly reduces the daily mental burden on patients by automating complex metabolic decision-making.

B. Research Question(S), Hypotheses, Engineering Goal(S), Expected Outcomes:

This study, titled "GLUCO-MATH: A Predictive Algorithmic System Utilizing Non-Linear Differential Models of Subcutaneous Insulin Absorption Integrated with Fuzzy Logic for Proactive Hypoglycemic Mitigation," aims to develop and validate a hybrid closed-loop controller that improves glycemic management by handling the chaotic and uncertain nature of human metabolism.

Specifically, it seeks to answer the following questions:

1. What is the comparative performance of the Non-Linear Differential Equation (NLDE) model "GLUCO-MATH" model versus traditional linear models in its ability to accurately capture metabolic oscillations and minimize glycemic variability across diverse scenarios?
2. What is the effectiveness of a Fuzzy Logic Controller in translating vague patient inputs (e.g., "high stress," "low activity") into precise insulin delivery adjustments to reduce glycemic variability?
3. Is there a significant difference in the frequency of hypoglycemic events and Time-in-Range (TIR) between the proposed hybrid controller and traditional reactive controllers during in silico testing?

It was hypothesized that:

1. The developed NLDE model will demonstrate lower Root Mean Square Error (RMSE) in predicting glucose trends following complex physiological disturbances (meals and exercise) compared to standard linear models like the Bergman Minimal Model.

2. The Fuzzy Logic Controller will significantly reduce the required accuracy of carbohydrate counting inputs while maintaining or improving the patient's Time-in-Range (TIR) compared to systems requiring precise numerical inputs.
3. The proactive 60-minute prediction window provided by the hybrid controller will decrease the frequency of nocturnal hypoglycemic events by over 50% in silico simulations compared to reactive PID-based controllers.

The engineering goal of this study is to design, simulate, and validate a hybrid algorithmic framework (software) that processes real-time Continuous Glucose Monitor (CGM) data and subjective user inputs to autonomously control insulin pump delivery, optimizing for safety and hypoglycemia prevention.

The expected outcome of this study is to produce a functional, validated Python-based simulation (In Silico) that demonstrates superior glycemic control, higher Time-in-Range, and reduced hypoglycemia compared to existing linear models, thereby proving the viability of a hybrid approach for a next-generation artificial pancreas.

C. Materials and Methods

a. List of Materials

The materials that will be used are Python programming environment, SciPy and scikit-fuzzy libraries, py-mgipsim simulator, Nightscout cloud platform, and Abbott FreeStyle Libre 3 sensors. Python will be used as the primary software environment because it is the most flexible and free option for developing the GLUCO-MATH framework. SciPy and scikit-fuzzy will be utilized to solve the Non-Linear Differential Equations (NLDE) and build the Fuzzy Logic Controller (FLC) layers that process "vague" human variables. To validate the algorithm safely, the py-mgipsim library will be used to simulate virtual patients with Type 1 Diabetes, acting as "digital twins" for testing. Nightscout will serve as the essential data bridge, allowing the GLUCO-MATH code to "pull" real-time glucose data through an API. Finally, Abbott FreeStyle Libre 3 sensors will be the hardware source for glucose data, as they are classified as integrated CGMs (iCGM) designed to be compatible with third-party software.

b. Procedures

1. Conceptualization of the Algorithmic Framework

The software architecture will be designed to merge Non-Linear Differential Equations (NLDE) with Fuzzy Logic to address the chaotic nature of human metabolism. The mathematical layout will focus on creating a "digital twin" capable of modeling non-linear delays in subcutaneous insulin absorption and metabolic oscillations. This step involves defining the fuzzy membership functions that will eventually translate linguistic inputs like "high stress" or "moderate exercise" into precise mathematical adjustments.

2. Gathering of Software and Data Tools

The primary development environment will be set up by installing Python and its essential scientific libraries, specifically SciPy for numerical integration and scikit-fuzzy for the inference engine. Access to the Nightscout API and Tidepool will be established to act as a computational "middleman" for pulling real-time or historical CGM data. The py-mgipsim library will also be downloaded to serve as the environment for virtual patient testing.

3. Programming and Algorithmic Development

The GLUCO-MATH framework will be coded using Python, where the SciPy library will be programmed to solve the Non-Linear Differential Equations. The scikit-fuzzy library will then be used to construct the logic layers that process "vague" human variables to refine insulin delivery. The code will be structured to produce a visual Prediction Horizon, forecasting blood sugar trends 30-60 minutes in advance to allow for proactive rather than reactive management.

4. Testing and Debugging (In Silico)

The algorithm will undergo rigorous in silico testing using the py-mgipsim simulator to detect bugs or errors in the predictive model. This simulation environment allows for the testing of "digital twins" of Type 1 Diabetes patients without physiological risk. The code will be debugged to ensure that the 60-minute prediction window accurately identifies potential hypoglycemic events before they occur.

5. Data Gathering and Validation

Data will be gathered by running simulations across three distinct metabolic scenarios: low-stress with accurate carbohydrate counting, high-stress with estimated counting, and physical activity with manual overrides. The performance of the system will be evaluated using the Root Mean Square Error (RMSE) to quantify the gap between predicted blood glucose levels and actual sensor data from the iCGM. Finally, a One-Way ANOVA will be conducted to determine if there are statistically significant differences in the Time-in-Range (TIR) achieved across the different simulated conditions.

1. Accuracy Rate Test (RMSE)

This test aims to determine the predictive precision of the NLDE model by calculating the residual difference between forecasted and observed glucose values. A lower RMSE value will serve as mathematical proof that the non-linear model successfully replicates unique metabolic oscillations.

2. Functionality Rate Test (TIR)

The effectiveness of the Fuzzy Logic Controller will be determined by measuring the patient's Time-in-Range (TIR) during the simulation. This test evaluates if the system can maintain safety and glucose stability even when provided with "fuzzy" or imprecise carbohydrate inputs.

3. Simulation Test (Hypoglycemic Mitigation)

The test aims to determine if the proactive prediction window can decrease the frequency of nocturnal hypoglycemic events. The system should autonomously adjust insulin delivery to mitigate hypoglycemia at least 30-60 minutes before it is projected to occur in the virtual patient.

c. Risk and Safety

The GLUCO-MATH research will involve the collection and analysis of human physiological data obtained from glucose sensors, which will subsequently be processed through the py-mgipsim simulation framework. Although the study will not involve direct physical intervention or clinical treatment, the primary risks will include potential breaches of data privacy and the risk of participants misinterpreting experimental predictions as medical advice. To ensure the safety and confidentiality of all involved, all sensor data will be strictly de-identified and encrypted prior to being integrated into the simulator, ensuring that no personally identifiable information (PII) will be accessible during the research process. Furthermore, to prevent any risk of medical mismanagement, it will be clearly communicated to participants that the GLUCO-MATH system will be a predictive computational model and not a diagnostic tool. The results generated by the algorithm will be for research purposes only and must not be used to adjust medication or insulin dosages. Participants will be instructed to continue following the protocols of their certified medical devices and healthcare providers, treating the system’s outputs as secondary, non-clinical data. All computational procedures will be monitored for technical errors to ensure that the data analysis remains accurate and does not lead to erroneous health-related conclusions.

d. Data Analysis

The predictive accuracy of the Non-Linear Differential Equation (NLDE) model will be computed by determining the Root Mean Square Error (RMSE) between the predicted glucose values and the actual CGM readings using the formula:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(G_i^{\text{actual}} - G_i^{\text{predicted}}\right)^2}$$

Where:

n = total number of time points

G_i^{actual} = actual CGM glucose at time i

$G_i^{\text{predicted}}$ = predicted glucose at time i

TABLE 1. Predictive accuracy rate of the glucose model

Variable	GLUCO-MATH (NLDE)
Total Number of Time Points (n)	
Sum of Squared Errors	
RMSE	

The functionality rate will be computed by getting the percentage of the number of times the glucose value stayed within the safe range (70–180 mg/dL) when the system was operating, given by the formula:

$$\text{Functionality Rate (\%)} = \frac{\text{Number of Readings Within Range}}{\text{Total Number of Readings}} \times 100$$

TABLE 2. Functionality rate of the GLUCO-MATH system

Variable	TIR
Total Number of Success Trials	
Total Number of Trials	30
Percentage	

The hypoglycemia mitigation rate will be computed by determining the percentage reduction in hypoglycemic events after GLUCO-MATH intervention using the formula:

$$\text{Mitigation Rate (\%)} = \frac{H_{baseline} - H_{GLUCO}}{H_{baseline}} \times 100$$

Where:

$H_{baseline}$ = hypoglycemic events using linear controller

H_{GLUCO} = hypoglycemic events using GLUCO-MATH

TABLE 3. Hypoglycemia mitigation rate

Variable	Hypoglycemia
Baseline Hypoglycemic vents	
Post-Adjustment Events	
Mitigation Percentage	

To determine if GLUCO-MATH performs consistently under different metabolic conditions, results will be recorded across three scenarios.

TABLE 4. Scenario performance summary

Scenario	Mean RMSE	Mean TIR (%)	Hypoglycemia Events
Low stress			
High stress			
Physical Activity			

Mean RMSE Formula:

$$\text{Mean RMSE} = \frac{1}{\text{m}} \sum_{j=1}^{m} RMSE_j$$

Where:

m = number of subjects or simulation runs

$RMSE_j$ = RMSE of subject j

Mean TIR Formula:

$$\text{Mean TIR (\%)} = \frac{1}{\text{m}} \sum_{j=1}^{m} TIR_j$$

Where:

m = number of subjects or simulation runs

TIR_j = TIR of subject j

A One-Way ANOVA will be conducted to determine if there is a significant difference in TIR across the three scenarios.

If $p < 0.05$, a Tukey HSD post-hoc test will be used to identify which scenario shows the greatest improvement under GLUCO-MATH.

e. Proposed Tables

TABLE 5. Model Comparison

Model	Scenario	Mean RMSE	Mean TIR	Hypoglycemia Events
Linear (PID/MPC)				
Non-Linear (GLUCO-MATH)				

TABLE 6. Effectiveness of the Fuzzy Logic Controller

Subject ID	Scenario (Fuzzy Linguistic Input)	Insulin Adjustment (U/hr)	Pre-Adjustment Glucose (mg/dL)	Post-Adjustment Glucose (mg/dL)	TIR (%)
VP01					
VP02					
VP03					

TABLE 7. TIR and Hypoglycemia Comparison Between Controllers

Subject ID	Scenario	Controller Type	TIR (%)	Hypoglycemia Events
VP01				
VP02				
VP03				

Bibliography

a. Journal Articles

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Razak, Tajul Rosli, Ahmad Zia Ul-Saufie, Mohamad Hanis Yusoff, Mohammad Hafiz Ismail, Shukor Sanim Mohd Fauzi, and Nurul Ain Mohd Zaki. "Python scikit-fuzzy: Developing a Fuzzy Expert System for Diabetes Diagnosis." *IAES International Journal of Artificial Intelligence (IJ-AI)* 13, no. 2 (2024): 2197–2206. https://doi.org/10.11591/ijai.v13.i2.pp2197-2206.

b. Medical Reference Chapters

Sapra, Amit, and Priyanka Bhandari. "Diabetes." In *StatPearls*. Treasure Island, FL: StatPearls Publishing, 2023. https://www.ncbi.nlm.nih.gov/books/NBK551501/.

c. Government and Global Health Publications

World Health Organization. "Diabetes." Fact sheets. April 5, 2024. https://www.who.int/news-room/fact-sheets/detail/diabetes.

Gantt Chart

Research Title: GLUCO-MATH: A Predictive Algorithmic System Utilizing Non-Linear Differential Models of Subcutaneous Insulin Absorption Integrated with Fuzzy Logic for Proactive Hypoglycemic Mitigation

Research Proponents: Jeycee Babes O. Manalo, Alliah Joyce E. Ciria, & Danielle Kelly G. dela Torre

	2026											
Activities	January	February	March	April	May	June	July	August	September	October	November	December
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Legend:

A. Conceptualization of Research Title

B. Research Proposal

C. Composition and Formulation of the Research Plan

D. Coordination with Consultant

E. Gathering of Software and Data Tools (Python, SciPy, scikit-fuzzy)

F. Configuration of Nightscout API and Tidepool Data Bridge

G. Programming and Algorithmic Development of NLDE

H. Construction of Fuzzy Logic Controller for Linguistic Inputs

I. Development of Visual Prediction Horizon Forecasting

J. Testing and Debugging (In Silico) using py-mgipsim

K. Gata Gathering

L. Data Analysis

M. Composition, Revision, and Refinement of Research Paper for DSTF

N. Composition, Revision, and Refinement of Paraphernalia for DSTF